

CompTIA.

# A+ Core 2

Exam 220-1202

A+ 220-1202

# Module 15



## Securing Windows

## Prepare for A+ Core 2 by:

- Configuring Windows networking
- Troubleshooting Windows networking
- Configuring Windows security settings
- Managing Windows shares

# Lesson 15.1

## Logical Security Concepts

# Logical Security Controls

## IAM

- Identification, Authentication, Authorization, Access Control

## Access control lists

## Implicit deny

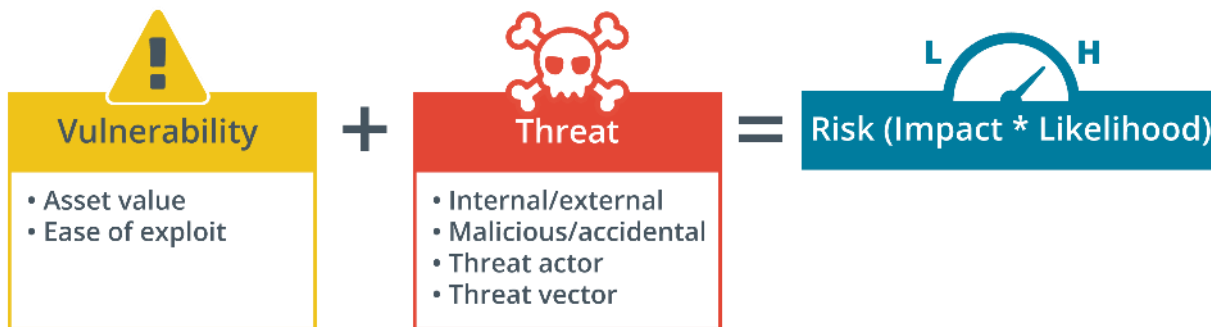
- Deny everything unless a rule says to allow it

## Least privilege

- Provide the right access for each user or system
- Nothing more, nothing less

# Information Security

- Vulnerability
  - Any weakness that, if exploited, could cause damage or a security breach
- Threat
  - Potential to exploit a vulnerability and breach the system
- Risk
  - Likelihood and impact of threat actor exercising vulnerability



# Hashing and Encryption Concepts

- Symmetric
  - Single key
- Asymmetric
  - Key pair
- Hash
  - Function takes any amount of data and produces a fixed length output (digest)
  - Does NOT contain the actual data
  - Used primarily for integrity checks

# Digital Signatures and Key Exchange

## Digital signature

- Proves a message or digital certificate has not been altered or spoofed

## Key exchange

- Allows two systems to exchange a key used for encryption
- Asymmetric algorithm keys used to encrypt symmetric keys for the exchange

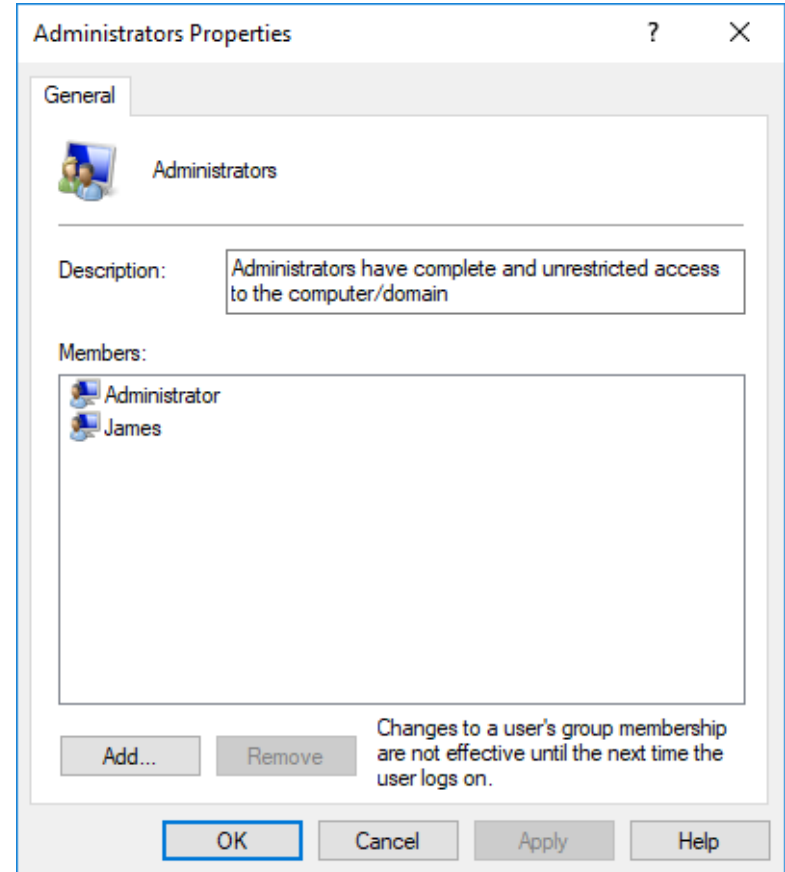


# User and Group Accounts (Slide 1 of 2)

- Local
  - Only used for accessing a single system or entity
- Microsoft account
  - Profile can sync through an online portal
- Security groups
  - Collection of accounts that will be assigned similar permissions
    - Administrator group
      - All files and objects in system
    - Guest group
      - Legacy group with same permissions as User group
    - Power user group
      - Between Users group and full administrator

# User and Group Accounts (Slide 2 of 2)

- Local Users and Groups
- net user commands
  - Manage accounts using command line utility

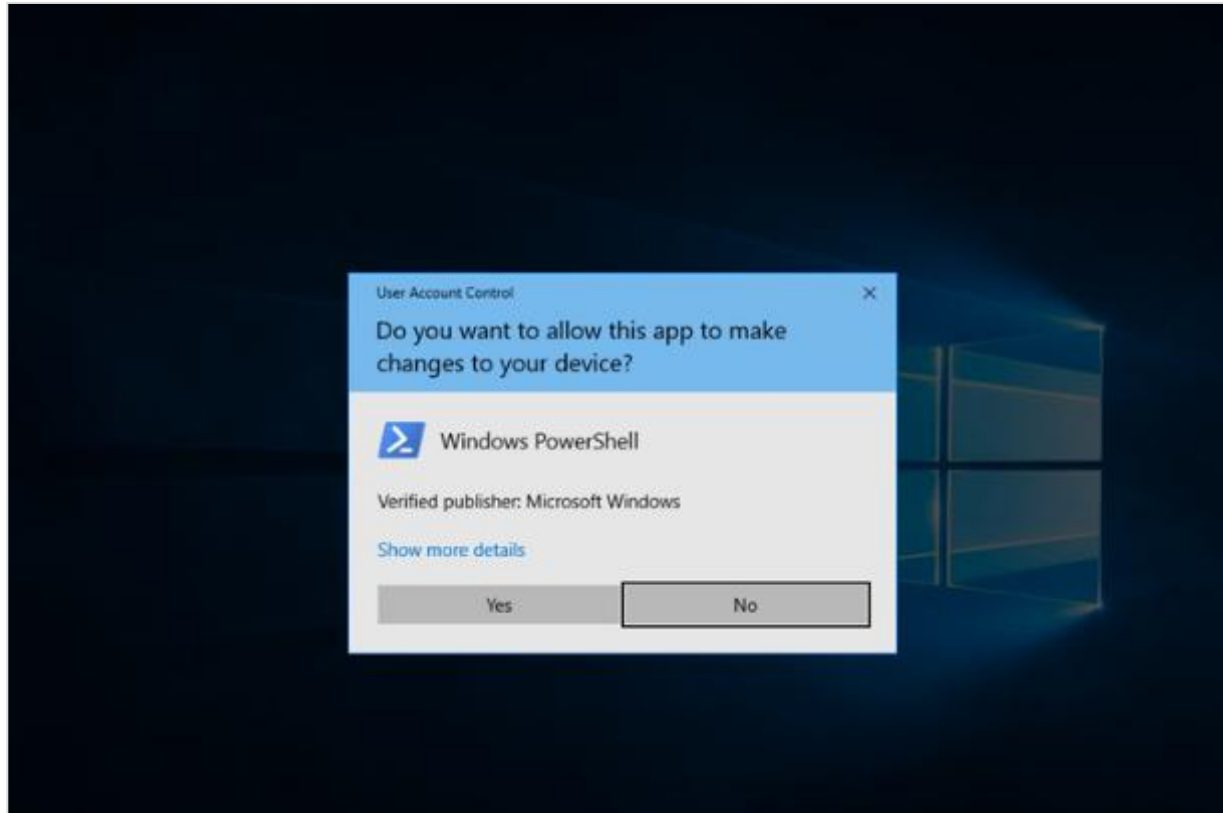


*Screenshot courtesy of Microsoft.*

# User Account Control (Slide 1 of 3)

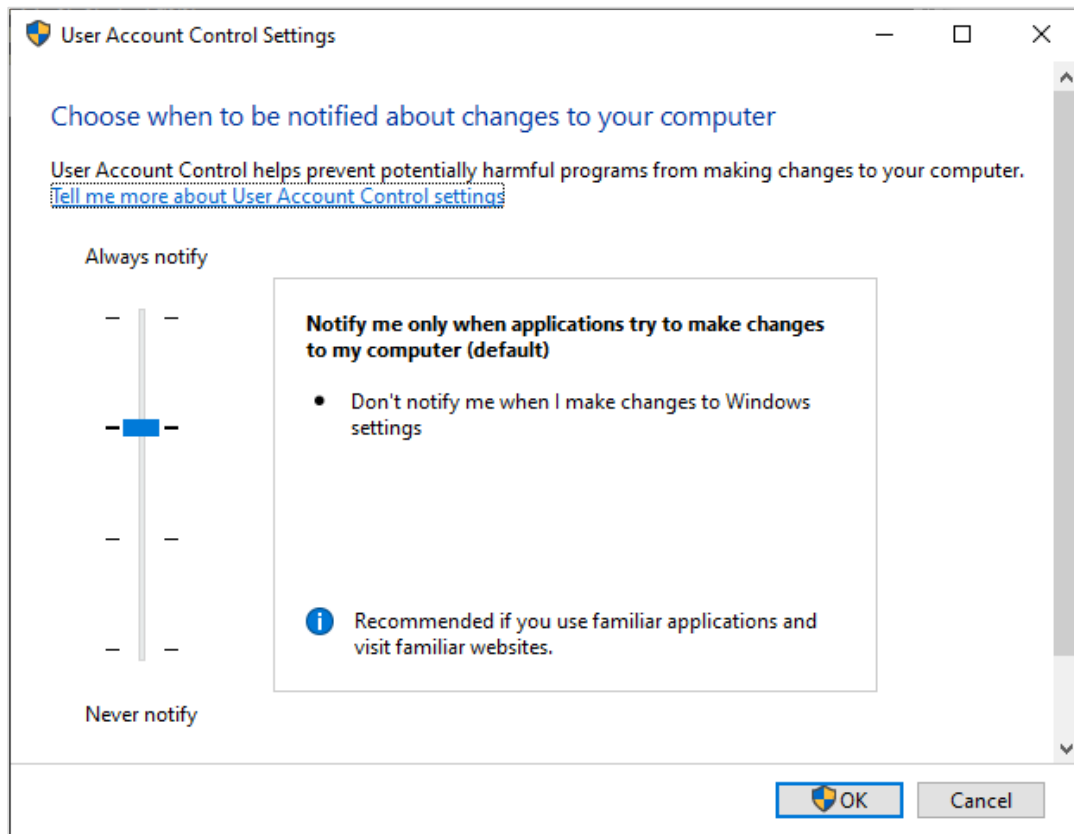
- Just-in-Time access
- Privileged Access Management
  - UAC provides implementation of JIT and PAM in Windows
  - User must confirm UAC pop up for change to be made

# User Account Control (Slide 2 of 3)



*Screenshot courtesy of Microsoft.*

# User Account Control (Slide 3 of 3)



*Screenshot courtesy of Microsoft.*

# Authentication Methods

- Proving you are what you say you are
- Zero trust framework
  - No automatic trust of user or device
- Multifactor authentication
  - Something you know, something you have, something you are, something you do, somewhere you are/aren't
- 2-step verification
- One-time passwords
- Authenticator applications
- Hard token authentication

## Lab Activity

- Lab: Create User Accounts  
(available in CertMaster Learn and CertMaster Perform)

## Lesson Review: Logical Security Concepts

- Logical Security Controls
- Information Security
- Hashing and Encryption Concepts
- Digital Signatures and Key Exchange
- User and Group Accounts
- User Account Control
- Authentication Methods



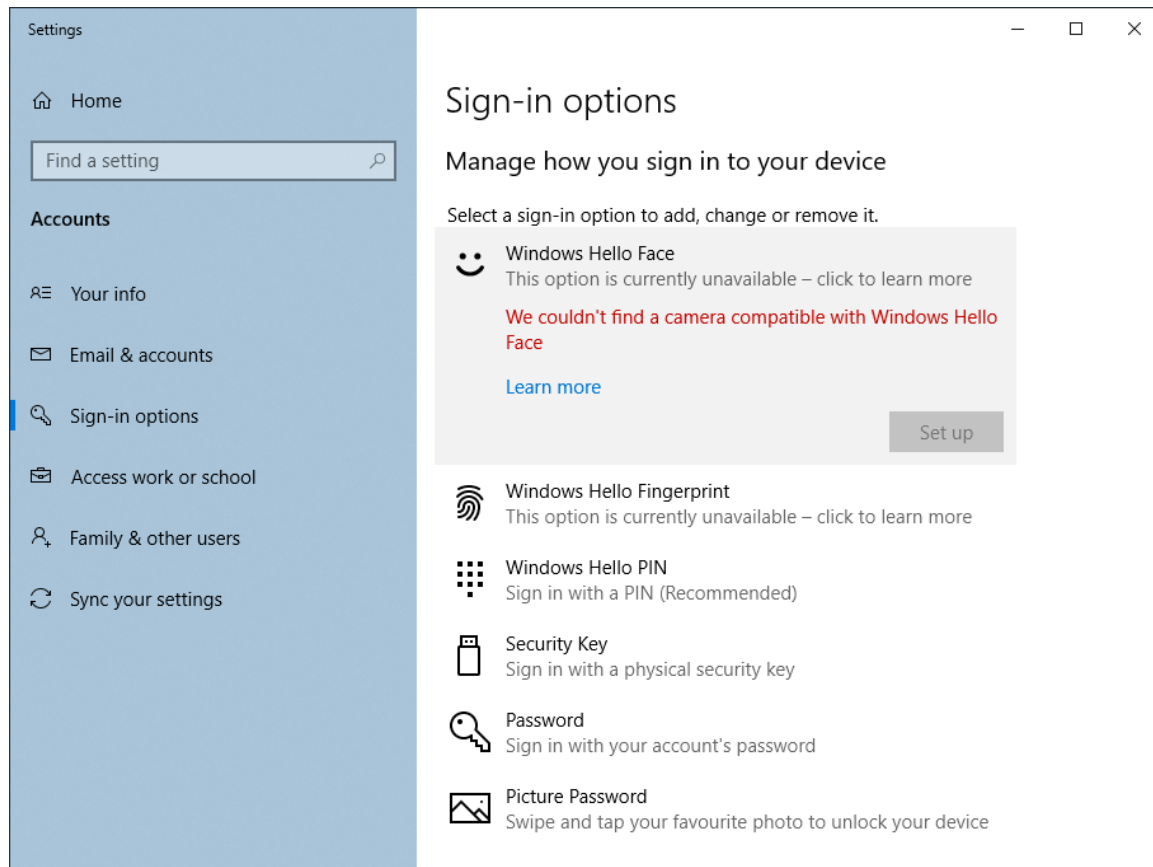
# Lesson 15.2

## Windows Security Settings

# Windows Login Options (Slide 1 of 2)

- Local, Windows network, remote login
- Username and password
- Windows Hello
  - Pin
  - Fingerprint
  - Facial recognition
  - Security key
- Single sign-on
- Security Assertions Markup Language (SAML)

# Windows Login Options (slide 2 of 2)

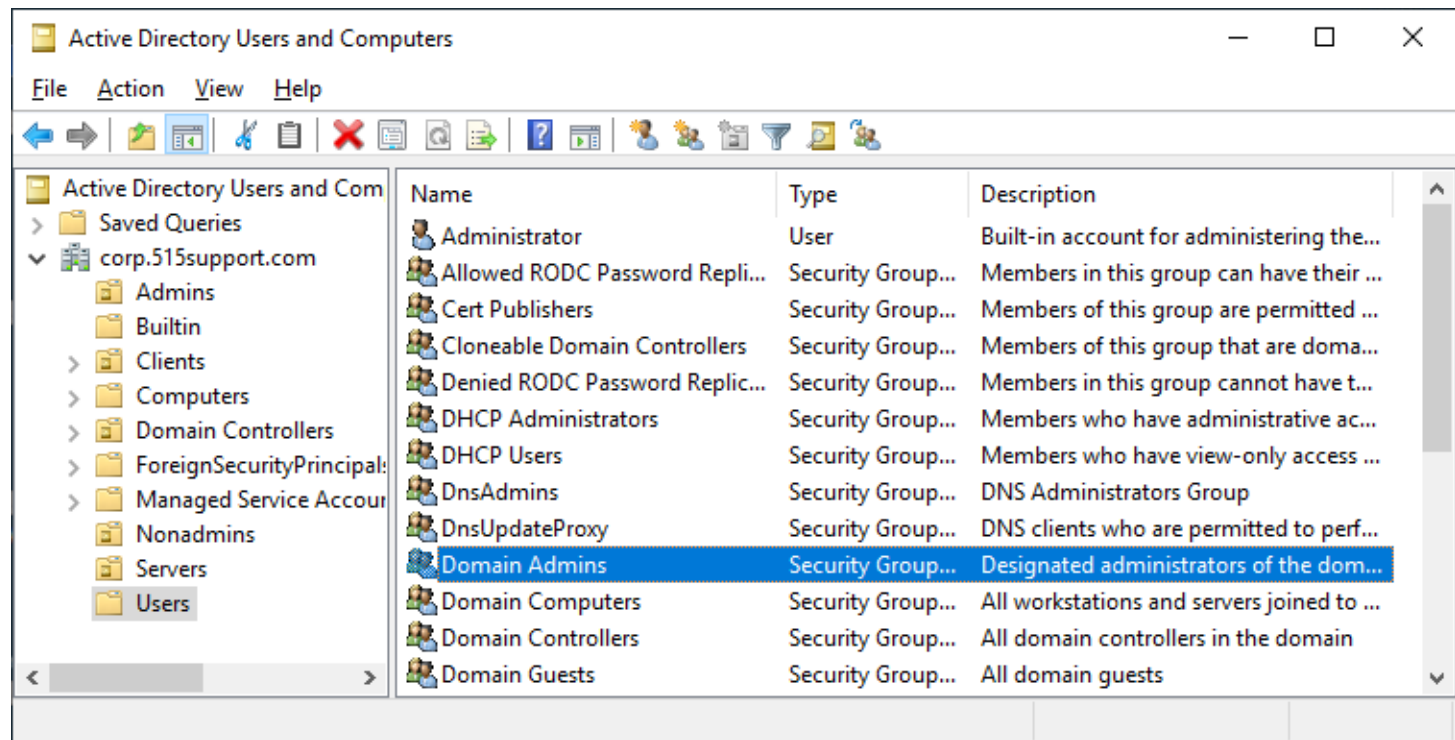


Screenshot courtesy of Microsoft.

# Windows Domains and Active Directory (Slide 1 of 2)

- Domain controllers
  - Server that stores a copy of the network information for Active Directory
- Member servers
  - Servers that are members of the domain but do not store a copy of Active Directory info
- Security groups
- Organizational units (OUs)

# Windows Domains and Active Directory (Slide 2 of 2)

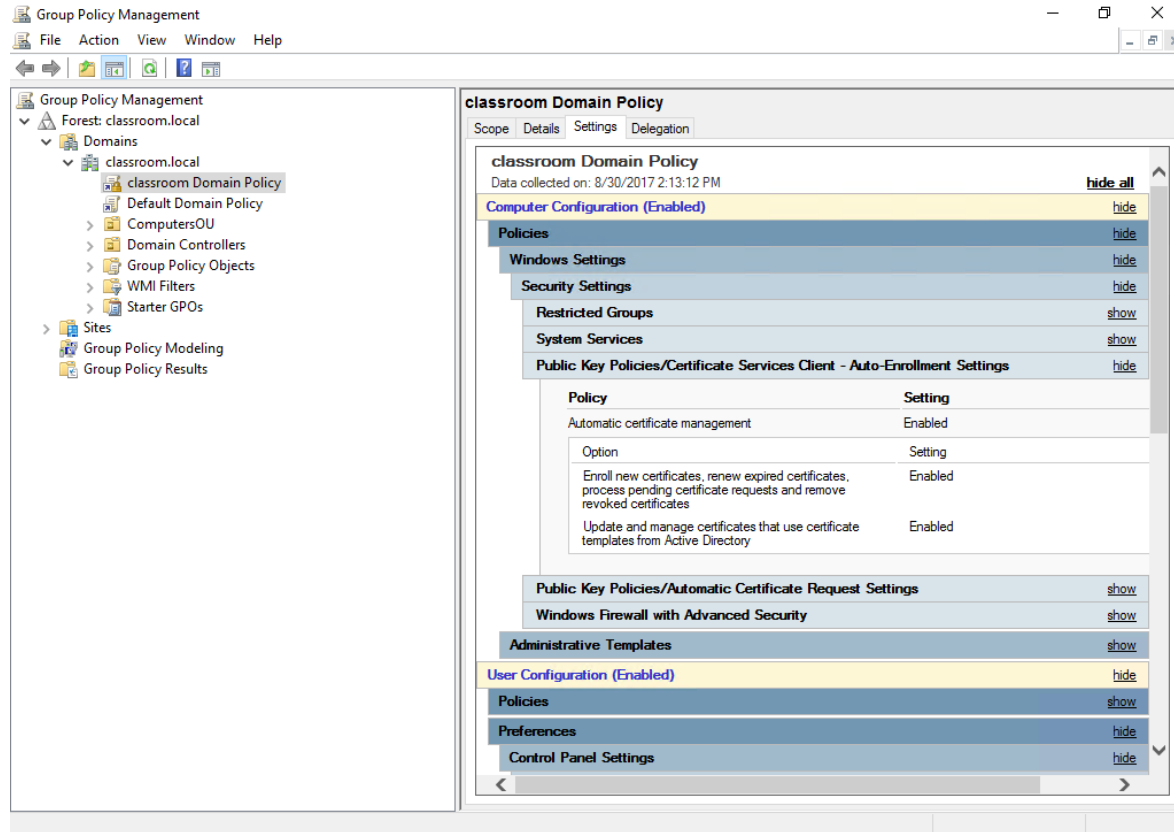


*Screenshot courtesy of Microsoft.*

# Group Policy and Login Scripts (Slide 1 of 2)

- Group Policy Management console
- Update policy
  - *gpupdate* command
  - *gpresult* command
- Login scripts

# Group Policy and Login Scripts (Slide 2 of 2)



Screenshot courtesy of Microsoft.

## Lab Activity

- Lab: Create OUs  
(available in CertMaster Learn and CertMaster Perform)
- Lab: Use System Commands  
(available in CertMaster Learn and CertMaster Perform)
- Lab: Group Policy Management  
(available in CertMaster Learn and CertMaster Perform)



## Lesson Review: Windows Security Settings

- Windows Login Options
- Windows Domains and Active Directory
- Group Policy and Login Scripts

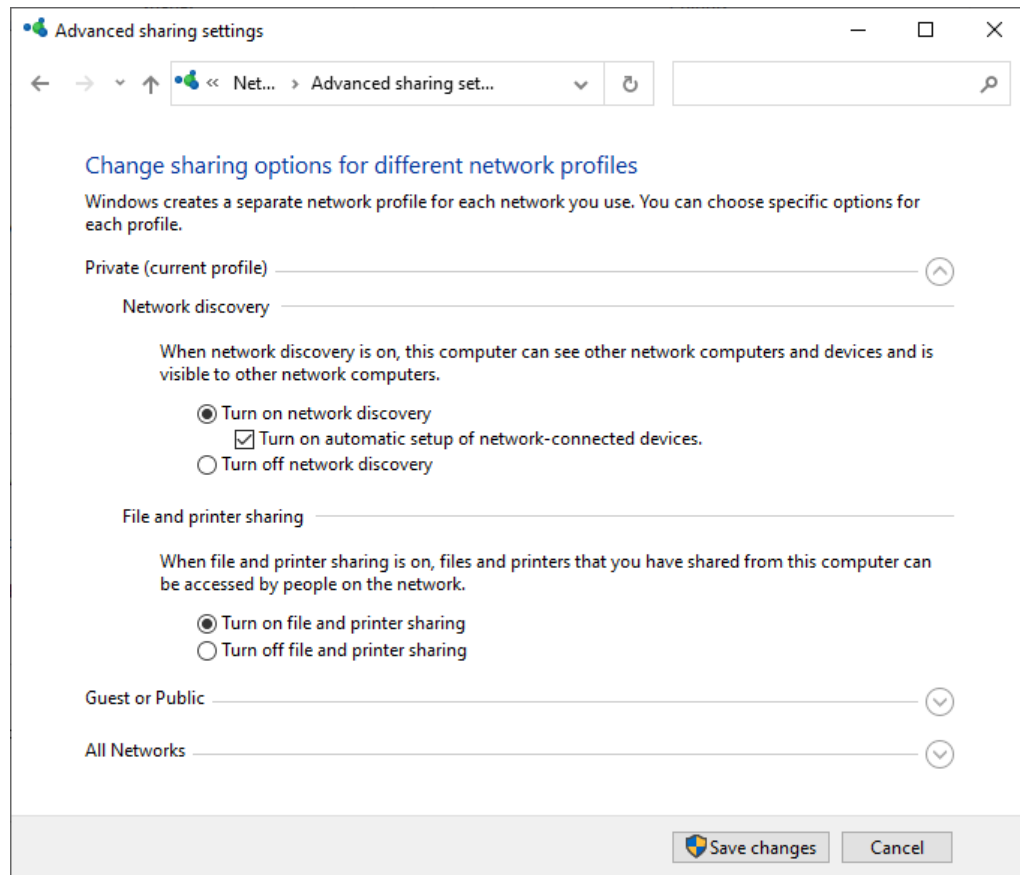
# Lesson 15.3

## Windows Shares

# Workgroup Setup (Slide 1 of 2)

- Join a workgroup
  - System Properties menu
- Network discovery and advanced sharing
  - File sharing
  - Printer sharing

# Workgroup Setup (Slide 2 of 2)

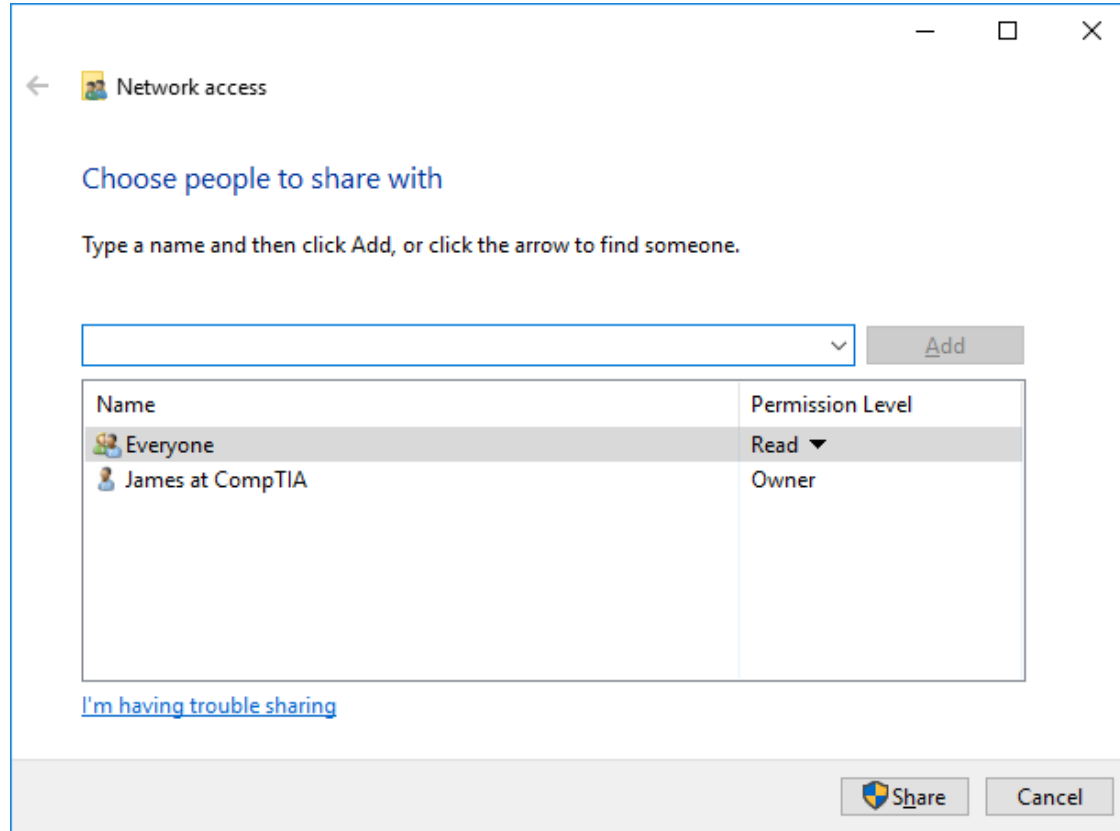


Screenshot courtesy of Microsoft.

# File Share Configuration (Slide 1 of 2)

- Right-click item and select Give access to
  - Select account, and then permissions
- Customize permissions
  - Properties menu
    - Share tab

# File Share Configuration (Slide 2 of 2)

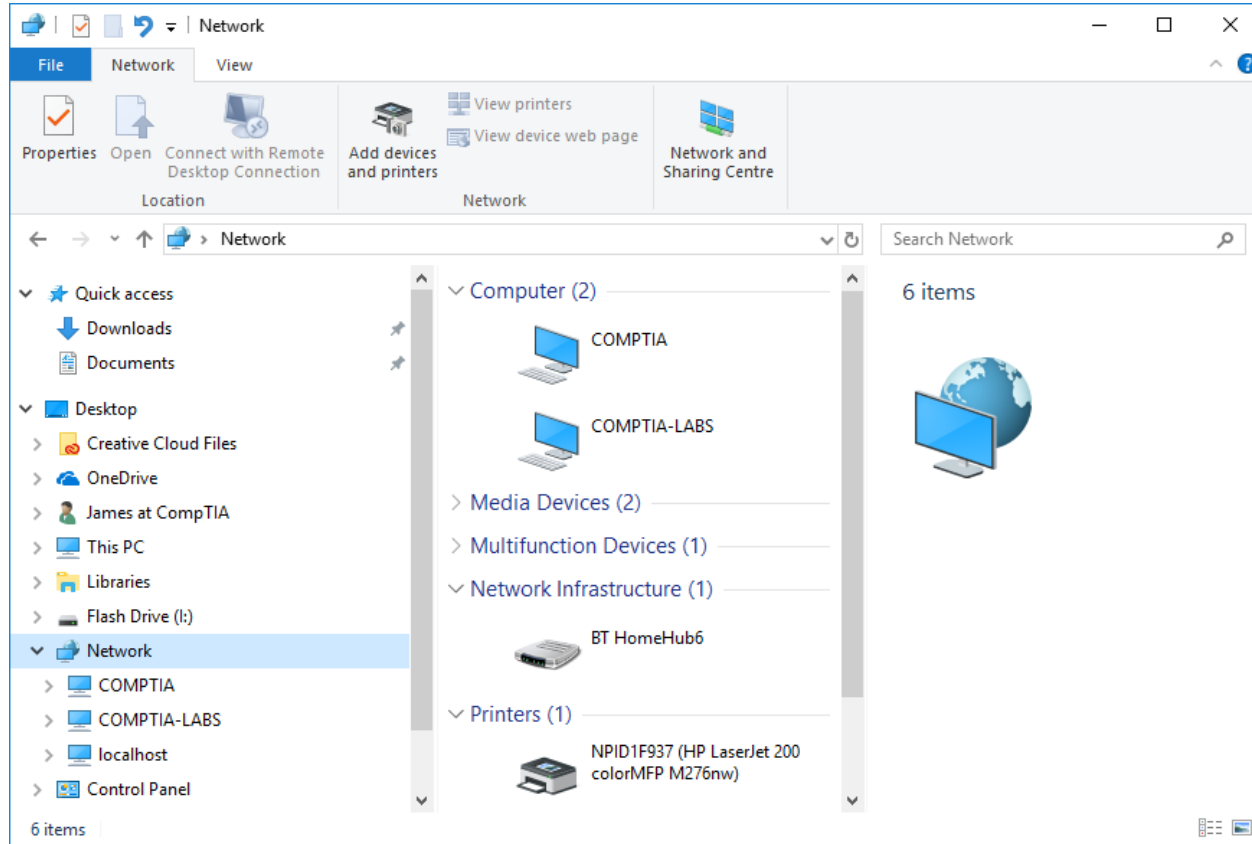


Screenshot courtesy of Microsoft.

# Network Browsing and Mapping Drives (Slide 1 of 3)

- Network object in File Explorer
  - Shows any networked device connected to the same network
- Mapped Drives
  - Network shares that can be reconnected every time a user logs in
- *net and net use* command

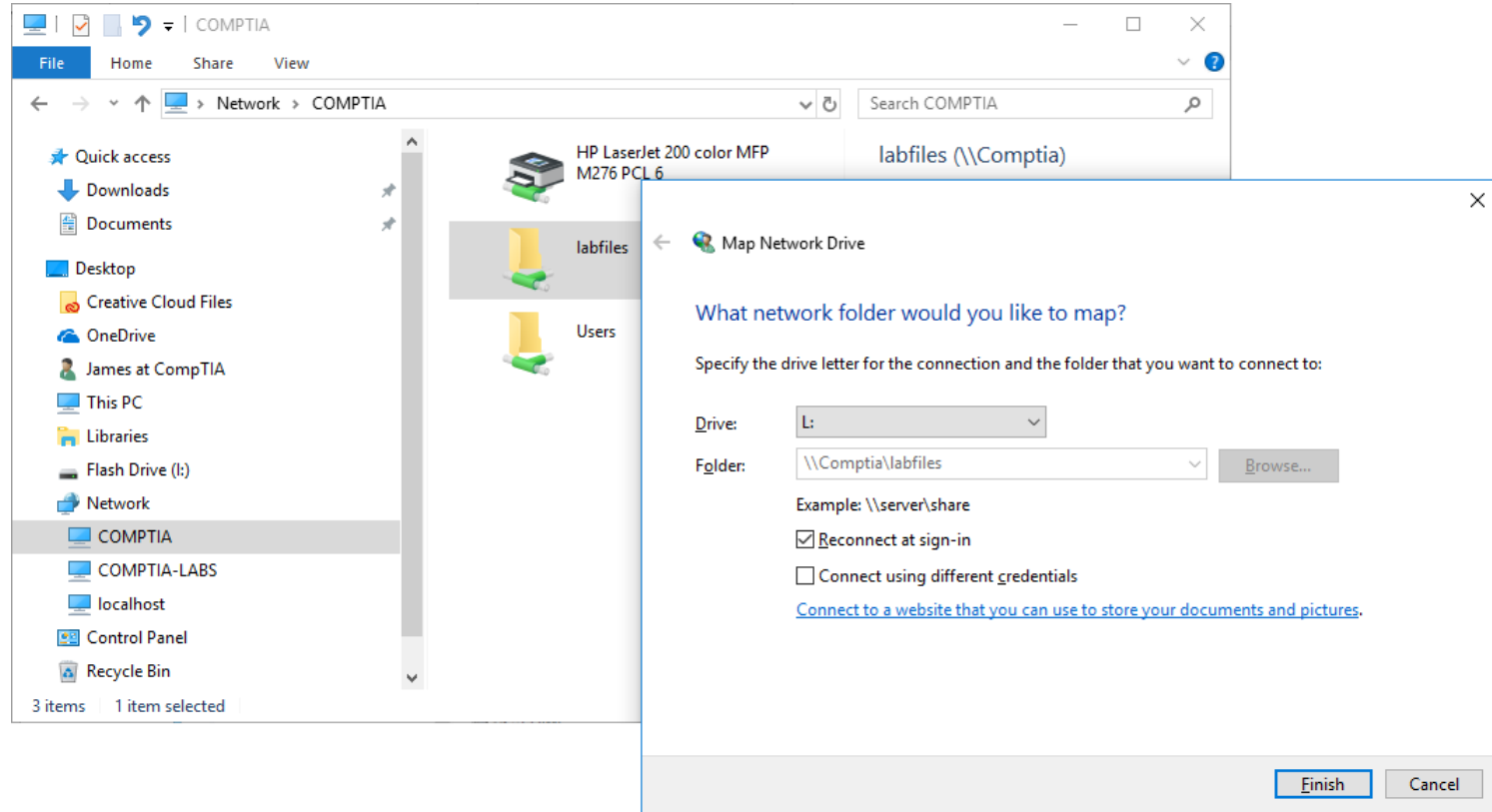
# Network Browsing and Mapping Drives (Slide 2 of 3)



*Screenshot courtesy of Microsoft.*

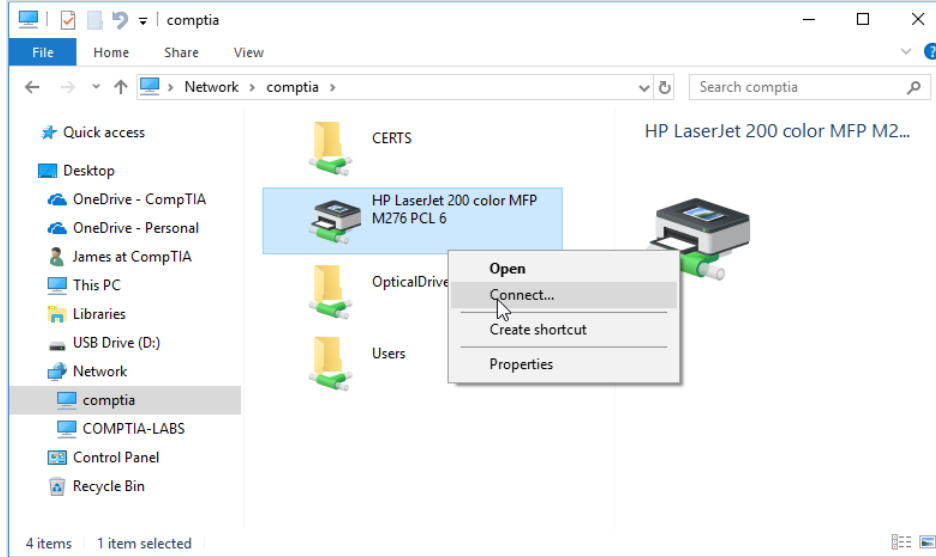


# Network Browsing and Mapping Drives (Slide 3 of 3)



*Screenshot courtesy of Microsoft.*

# Printer Sharing



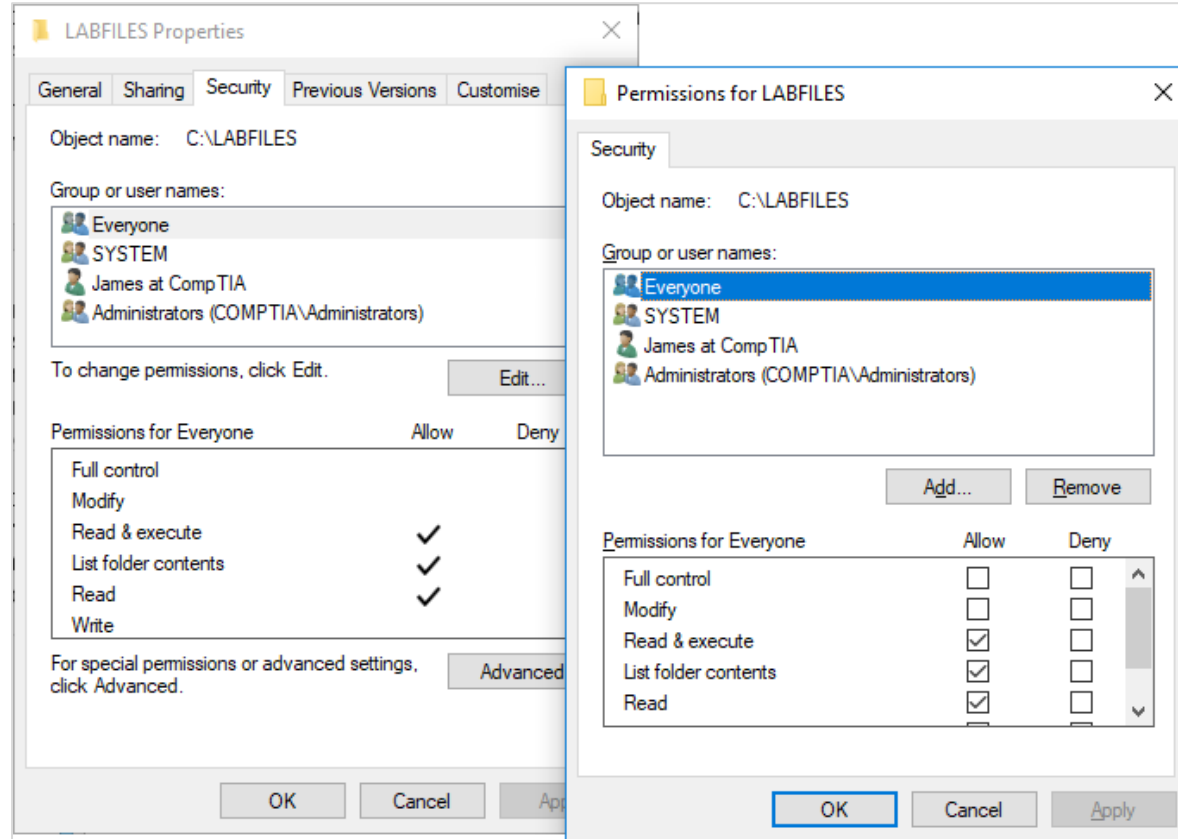
*Screenshot courtesy of Microsoft.*

- Printer Properties > Sharing tab
  - Additional drivers should be made available for different OSs that may connect to printer

# NTFS versus Share Permissions (Slide 1 of 2)

- Share permissions only apply when accessed over a network
- NTFS permissions are applied to the resource on local and network access
- NTFS permissions are set on Security tab of Properties dialog
  - Read, write, execute
- Explicit vs implicit deny or allow

# NTFS versus Share Permissions (Slide 2 of 2)



Screenshot courtesy of Microsoft.

# Permissions Inheritance

- Parent folder permissions trickle down to child subfolders and files
- Share permissions for across network access
  - NTFS permissions enforced for network and local access
- Most restrictive permission becomes effective permission
  - If one permission not allowed, the other allow, then not allow will be effective

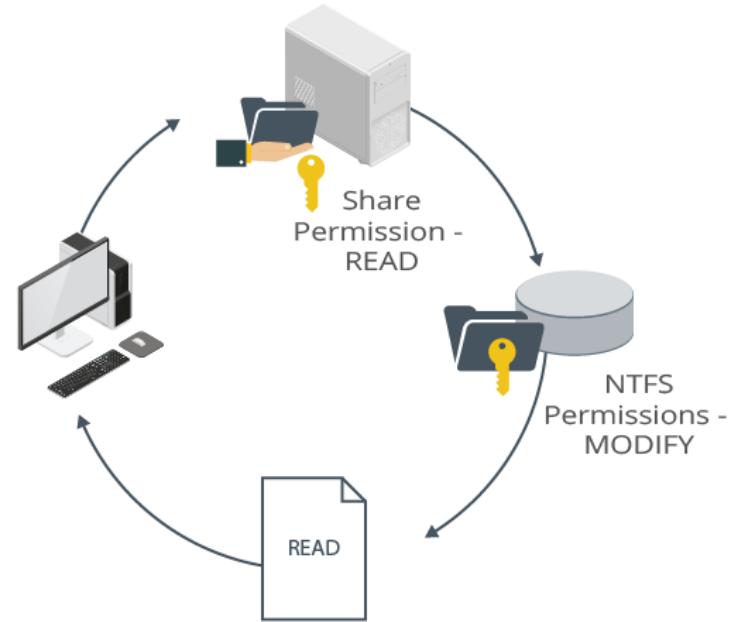
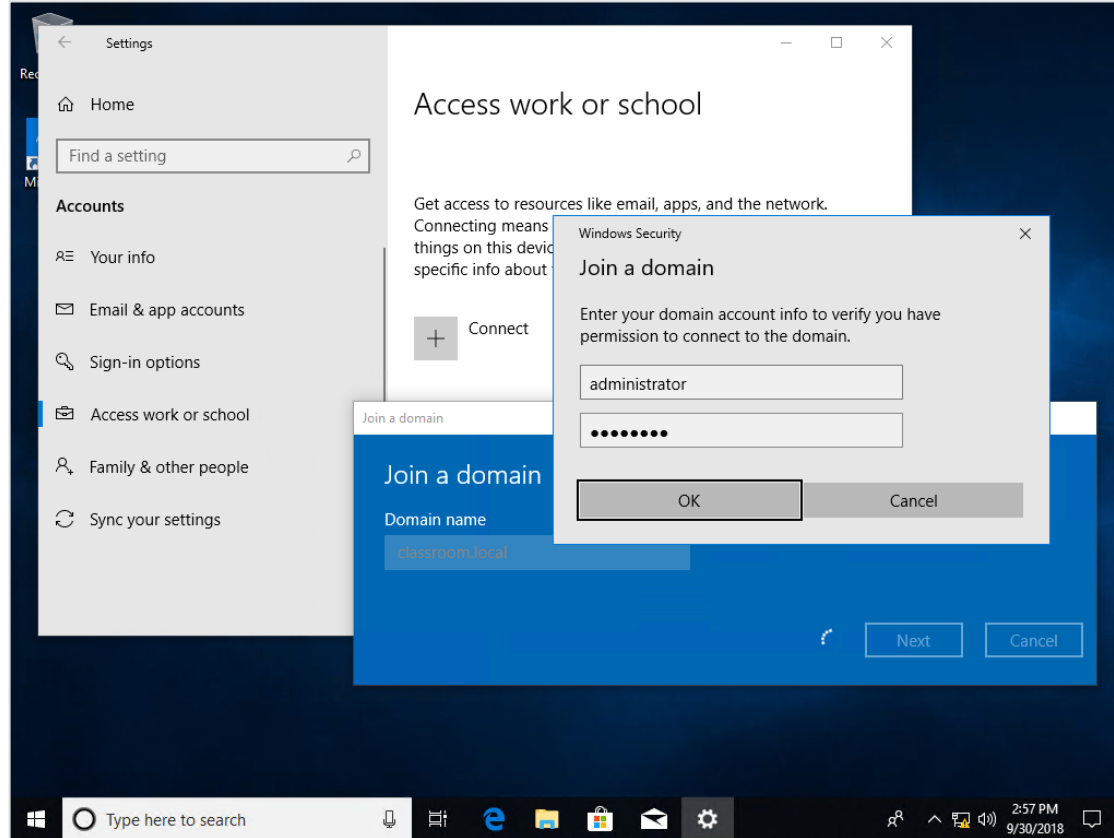


Image © 123RF.com.

## Domain Setup (Slide 1 of 3)

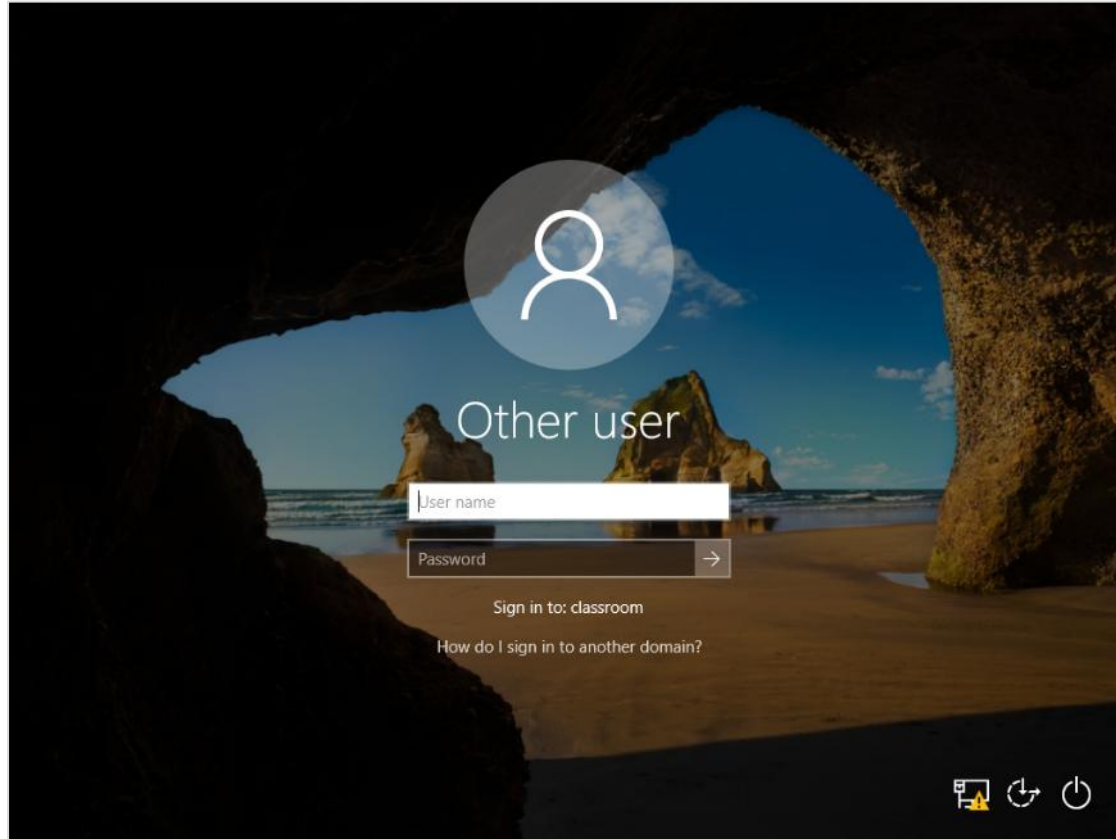
- Control and permissions set through group policy and domain security policy
- Account settings app
  - Access work or school option
- Can also use System Properties menu

# Domain Setup (Slide 2 of 3)



Screenshot courtesy of Microsoft.

# Domain Setup (Slide 3 of 3)



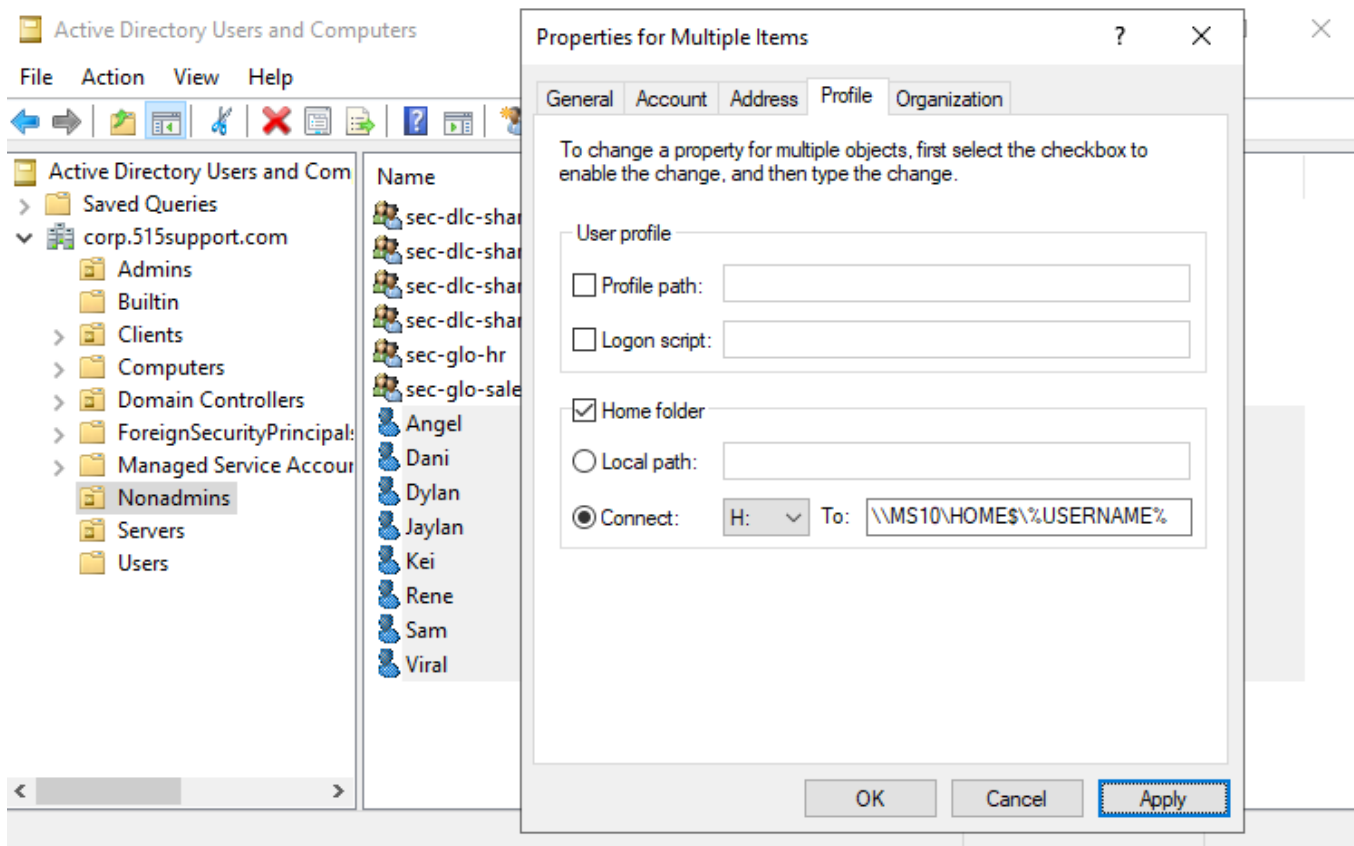
*Screenshot courtesy of Microsoft.*



# Home Folders (Slide 1 of 3)

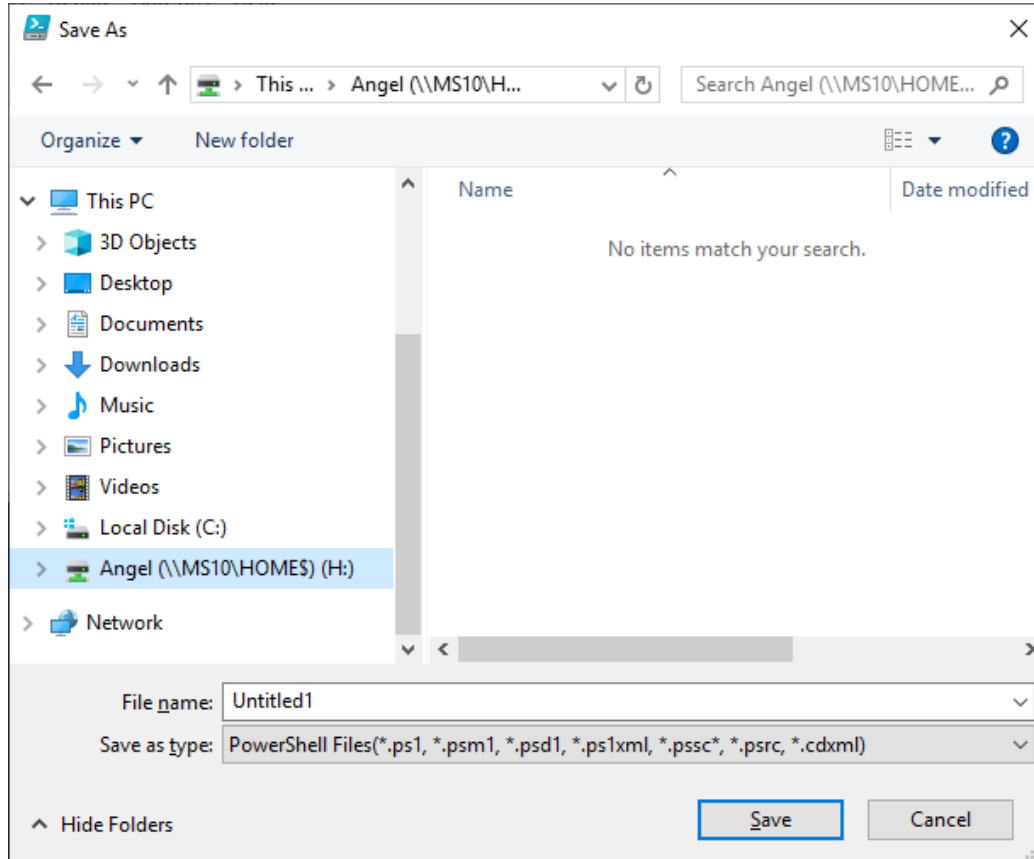
- Store files on server rather than local workstation
  - Centralized storage allows monitoring and regular back up
- Can be mapped using the profile settings of user account

# Home Folders (Slide 2 of 3)



Screenshot courtesy of Microsoft.

# Home Folders (Slide 3 of 3)

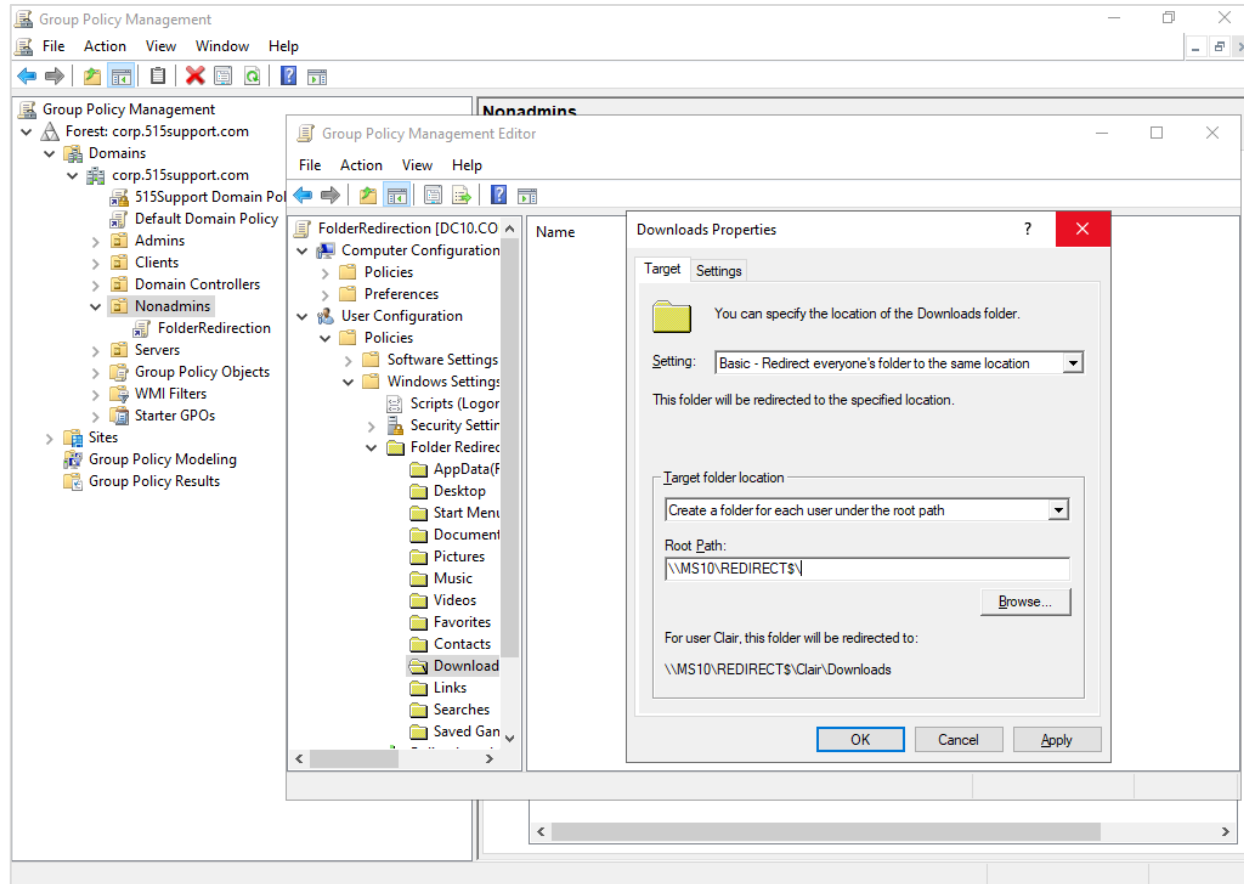


Screenshot courtesy of Microsoft.

# Roaming Profiles and Folder Redirection (Slide 1 of 2)

- Allows users to log in anywhere on network and have access to their files, applications, and settings
- Redirect profile folders to server-based storage for resiliency

# Roaming Profiles and Folder Redirection (Slide 2 of 2)



## Lab Activity

- Lab: Configure NTFS Permissions  
(available in CertMaster Learn and CertMaster Perform)
- Lab: Support AD Domain Networking  
(available in CertMaster Learn and CertMaster Perform)
- Challenge Live Lab: Manage Domain Accounts  
(available in CertMaster Perform and CertMaster Labs)

## Lesson Review: Windows Shares

- Workgroup Setup
- File Share Configuration
- Network Browsing and Mapping Drives
- Printer Sharing
- NTFS versus Share Permissions
- Permissions Inheritance
- Domain Setup
- Home Folders
- Roaming Profiles and Folder Redirection